

FACTSHEET FS07 · Size

Do smaller companies beat bigger ones?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Size premium (Fama-French SMB)	Banz (1981)	Size
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (smallest names in the universe)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

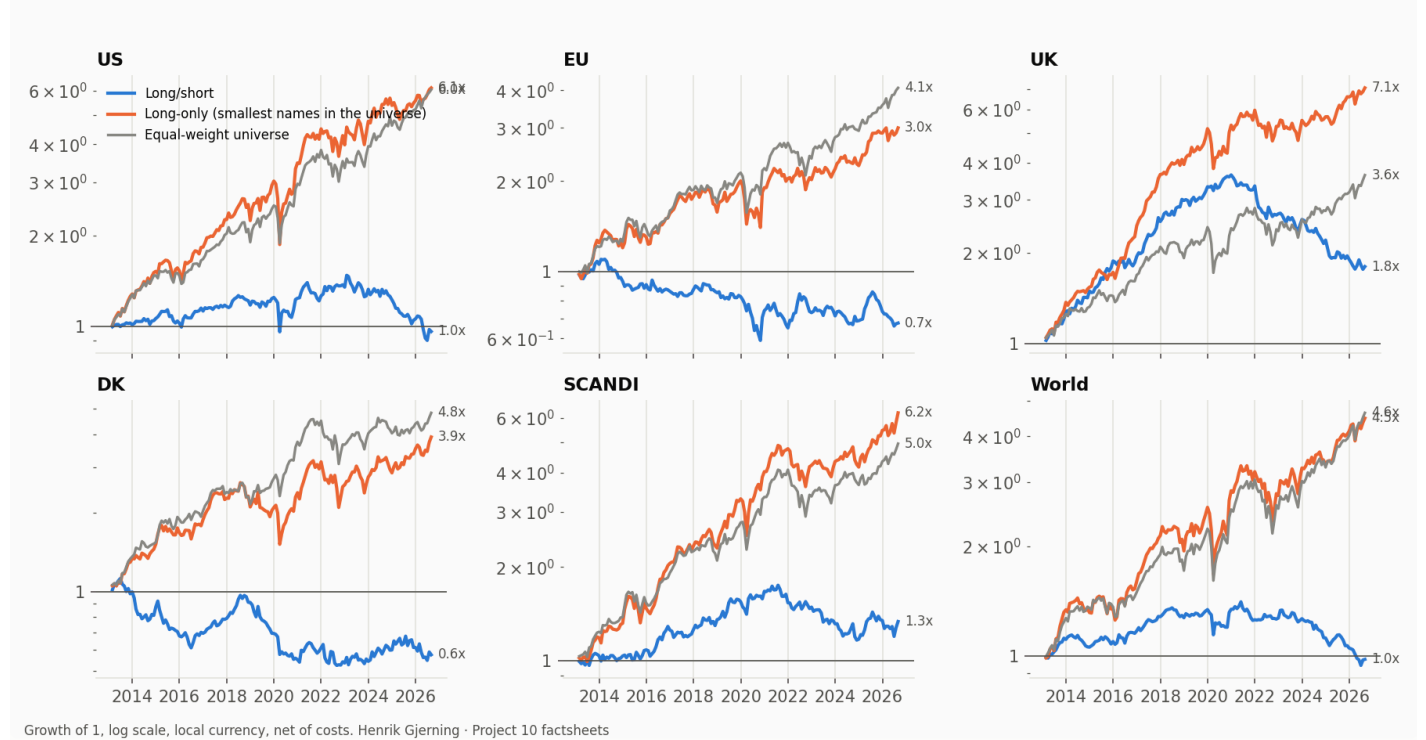
The claim. Small companies earn higher returns than large ones. Banz (1981) found it in US stocks 1936–1975; it became Fama & French's SMB factor, but has been weak since the early 1980s unless junk stocks are controlled for (Asness et al. 2018). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -3.0% to +4.9% a year and was positive in 4 of the six; no universe passes the 2.87 gate. The long-only book (smallest names in the universe) had a higher Sharpe than the equal-weight universe in 1 of the six (UK), with alphas of -2.6% to +5.8%, passing the gate in UK. **What goes wrong (§10):** the main drags are a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Size; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours Near 52-week high, 55-day breakout; after those neighbours and the market, nothing is left (alpha $|t| < 2$ everywhere): the signal is redundant.

1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long- only CAGR	Long- only Sharpe	Long- only max DD	EW CAGR	EW Sharpe	Size- weighted CAGR
US	503 / 10	+0.4%	0.17	-5.5% (-1.7)	0.41	+14.3%	0.77	-38%	+14.2%	0.96	+15.3%
EU	236 / 10	-2.2%	-0.64	-3.6% (-1.0)	0.12	+8.4%	0.57	-30%	+10.9%	0.78	+10.1%
UK	242 / 10	+4.9%	1.43	+2.9% (0.7)	0.20	+15.5%	1.06	-26%	+10.0%	0.74	+8.5%
DK	19 / 3	-3.0%	-0.79	-4.0% (-1.0)	0.08	+10.6%	0.64	-42%	+12.3%	0.82	+10.3%
SCANDI	62 / 5	+2.7%	1.07	+1.5% (0.6)	0.09	+14.4%	0.91	-26%	+12.5%	0.91	+10.7%
World	979 / 10	+0.2%	0.07	-3.1% (-1.5)	0.26	+11.7%	0.70	-34%	+12.0%	0.80	+10.8%

Monthly, local currency (World in USD). L/S = equal-weighted smallest names in the universe minus largest names, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Size on six stock universes, 2013-2026



2. Strategy description

Small companies earn higher returns than large ones. Banz (1981) found it in US stocks 1936–1975; it became Fama & French's SMB factor, but has been weak since the early 1980s unless junk stocks are controlled for (Asness et al. 2018).

Why it might work. Risk (distress, illiquidity; Fama & French 1993) or mispricing; much of the historical premium came from the smallest, least liquid stocks and January.

Rule	Original (Banz (1981))	This factsheet
Signal	Market capitalisation	US: market cap (Sharadar). Elsewhere: 63-day average traded value (price × volume), a proxy
Portfolio	Size deciles or SMB (2×3)	Extreme quantile L/S within the universe, monthly
Weighting	Value weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** No shares outstanding outside the US: size is proxied by traded value, which also measures liquidity and turnover.
- **Data gap:** The universes are blue chips; the classic size premium lives in micro and small caps, which are excluded by construction.

4. Signal creation

$SIZE_{i,t} = \log \text{market cap (US) or log 63-day average traded value (elsewhere) at month-end } t$; World ranks within each region first. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- Long/short** = smallest names in the universe minus largest names; **long-only** = smallest names in the universe; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 503 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+0.4%	-0.3%	11.7%	0.04 [-0.40, 0.50]	-0.03	-39%	50%	0.17	0.41	-5.5% (-1.7)	6.9%
Long-only (net)	+15.4%	+14.3%	20.0%	0.77 [0.37, 1.34]	1.14	-38%	65%	4.01	1.25	-2.6% (-1.3)	11.4%
Smallest names in the universe (gross)	+16.0%	+14.9%	20.0%	0.80 [0.39, 1.38]	1.20	-38%	65%	4.16	1.25	-2.1% (-1.0)	11.3%
Largest names (gross)	+14.7%	+14.6%	13.8%	1.06 [0.62, 1.60]	1.78	-25%	68%	5.37	0.84	+2.5% (1.8)	7.7%
EW universe	+14.5%	+14.2%	15.1%	0.96 [0.52, 1.53]	1.52	-25%	67%	4.79	1.00	–	8.9%
Size-weighted universe	+15.4%	+15.3%	14.4%	1.07 [0.61, 1.64]	1.79	-24%	69%	5.27	0.90	+2.3% (1.8)	8.3%

EU (11 national blue-chip indices, point-in-time), median 236 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-2.2%	-2.9%	11.9%	-0.18 [-0.76, 0.29]	-0.37	-46%	44%	-0.64	0.12	-3.6% (-1.0)	5.7%
Long-only (net)	+9.5%	+8.4%	16.6%	0.57 [0.13, 1.08]	0.85	-30%	59%	2.37	1.00	-2.0% (-0.8)	9.1%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Smallest names in the universe (gross)	+9.6%	+8.6%	16.6%	0.58 [0.14, 1.09]	0.87	-30%	59%	2.41	1.00	-1.8% (-0.8)	9.1%
Largest names (gross)	+11.3%	+10.7%	14.3%	0.79 [0.31, 1.36]	1.19	-28%	64%	3.24	0.87	+1.3% (0.6)	8.6%
EW universe	+11.4%	+10.9%	14.6%	0.78 [0.33, 1.33]	1.22	-26%	64%	3.33	1.00	—	8.4%
Size-weighted universe	+10.7%	+10.1%	14.3%	0.75 [0.28, 1.28]	1.14	-26%	60%	3.12	0.95	-0.1% (-0.1)	8.3%

UK (FTSE 350, point-in-time), median 242 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+4.9%	+4.4%	10.9%	0.45 [-0.24, 1.20]	0.65	-51%	53%	1.43	0.20	+2.9% (0.7)	5.6%
Long-only (net)	+15.6%	+15.5%	14.7%	1.06 [0.52, 1.81]	1.79	-26%	67%	4.32	0.93	+5.8% (3.0)	8.3%
Smallest names in the universe (gross)	+15.9%	+15.9%	14.7%	1.08 [0.54, 1.84]	1.84	-26%	67%	4.41	0.93	+6.1% (3.1)	8.3%
Largest names (gross)	+10.3%	+9.9%	13.2%	0.78 [0.23, 1.40]	1.19	-24%	63%	2.93	0.74	+2.6% (0.9)	7.5%
EW universe	+10.6%	+10.0%	14.2%	0.74 [0.26, 1.35]	1.08	-30%	59%	3.23	1.00	—	8.8%
Size-weighted universe	+9.0%	+8.5%	12.7%	0.71 [0.21, 1.31]	1.01	-27%	62%	2.94	0.82	+0.4% (0.2)	7.7%

DK (OMXC25, point-in-time), median 19 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-3.0%	-4.0%	14.4%	-0.21 [-0.71, 0.33]	-0.37	-53%	47%	-0.79	0.08	-4.0% (-1.0)	9.1%
Long-only (net)	+11.8%	+10.6%	18.3%	0.64 [0.13, 1.28]	0.85	-42%	58%	2.52	1.05	-1.7% (-0.6)	11.9%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Smallest names in the universe (gross)	+12.0%	+10.8%	18.3%	0.66 [0.14, 1.29]	0.87	-42%	58%	2.57	1.05	-1.5% (-0.6)	11.9%
Largest names (gross)	+14.5%	+13.7%	17.7%	0.82 [0.30, 1.39]	1.32	-29%	61%	3.18	0.97	+1.9% (0.8)	9.7%
EW universe	+12.9%	+12.3%	15.6%	0.82 [0.30, 1.44]	1.26	-33%	64%	3.21	1.00	—	9.3%
Size-weighted universe	+11.2%	+10.3%	16.6%	0.68 [0.11, 1.32]	0.94	-40%	61%	2.52	0.91	-0.6% (-0.2)	10.8%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 62 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.7%	+2.2%	10.7%	0.25 [-0.22, 0.72]	0.31	-33%	53%	1.07	0.09	+1.5% (0.6)	5.9%
Long-only (net)	+14.9%	+14.4%	16.4%	0.91 [0.42, 1.46]	1.47	-26%	60%	3.80	1.05	+1.4% (0.8)	9.4%
Smallest names in the universe (gross)	+15.0%	+14.5%	16.4%	0.91 [0.42, 1.47]	1.48	-26%	60%	3.82	1.05	+1.5% (0.9)	9.4%
Largest names (gross)	+11.8%	+11.2%	14.6%	0.80 [0.35, 1.35]	1.19	-26%	66%	3.59	0.96	-0.5% (-0.3)	9.2%
EW universe	+12.8%	+12.5%	14.1%	0.91 [0.39, 1.54]	1.42	-29%	67%	3.62	1.00	—	8.3%
Size-weighted universe	+11.2%	+10.7%	13.9%	0.80 [0.31, 1.37]	1.21	-27%	63%	3.26	0.95	-1.0% (-0.9)	8.6%

World (US + UK + EU, point-in-time), median 979 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+0.2%	-0.1%	7.6%	0.02 [-0.53, 0.56]	-0.03	-33%	50%	0.07	0.26	-3.1% (-1.5)	4.4%
Long-only (net)	+12.8%	+11.7%	18.2%	0.70 [0.25, 1.26]	1.02	-34%	63%	3.03	1.12	-1.3% (-1.0)	10.7%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Smallest names in the universe (gross)	+13.1%	+12.1%	18.2%	0.72 [0.27, 1.29]	1.06	-34%	63%	3.12	1.12	-1.0% (-0.7)	10.7%
Largest names (gross)	+12.2%	+11.9%	14.0%	0.87 [0.39, 1.46]	1.36	-27%	67%	3.80	0.86	+1.4% (1.2)	8.4%
EW universe	+12.6%	+12.0%	15.7%	0.80 [0.33, 1.38]	1.20	-29%	64%	3.53	1.00	–	9.4%
Size-weighted universe	+11.4%	+10.8%	15.0%	0.76 [0.29, 1.33]	1.14	-28%	63%	3.23	0.94	-0.4% (-0.5)	9.0%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+1.7%	+5.8%	+27.9%	-0.1%	+4.0%	+11.3%
2014	+4.7%	-12.1%	+16.8%	-12.4%	-0.1%	-4.7%
2015	-3.5%	-2.6%	+26.0%	-22.5%	+4.6%	+3.6%
2016	+9.7%	-5.0%	+3.5%	+2.3%	+15.7%	+7.0%
2017	+3.5%	+1.4%	+30.7%	+14.0%	+0.5%	+9.3%
2018	+1.1%	-2.5%	+17.5%	+15.1%	+7.5%	-0.6%
2019	+6.0%	-3.0%	+11.0%	-23.9%	+14.1%	+5.3%
2020	-0.6%	-10.1%	+9.5%	-20.6%	+8.0%	+0.2%
2021	+1.9%	-12.4%	-8.1%	+11.0%	-8.1%	-6.4%
2022	+6.8%	+12.4%	-19.2%	-12.5%	-3.1%	+1.1%
2023	-1.6%	-2.4%	-4.3%	+1.5%	-11.6%	-0.4%
2024	-7.7%	-3.6%	-14.6%	+10.5%	-10.7%	-11.4%
2025	-13.5%	+7.3%	-13.8%	+2.3%	+14.8%	-6.2%
2026	-9.2%	-8.5%	-5.3%	-6.3%	-0.8%	-6.8%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+28.2%	+24.6%	+37.0%	+34.1%	+18.7%	+34.1%
2014	+18.8%	-3.6%	+12.3%	+13.4%	+11.3%	-1.7%
2015	+0.3%	+8.5%	+12.1%	+10.5%	+11.6%	+2.0%
2016	+19.8%	+19.7%	+42.9%	+12.4%	+39.3%	+19.3%
2017	+23.2%	+14.2%	+48.2%	+20.8%	+13.8%	+33.5%

Year	US	EU	UK	DK	SCANDI	World
2018	-0.7%	-11.5%	+7.9%	-7.4%	-1.7%	-9.3%
2019	+35.0%	+27.0%	+31.6%	-0.5%	+43.8%	+31.6%
2020	+14.8%	-5.0%	+2.8%	+13.9%	+22.1%	+11.1%
2021	+29.5%	+7.3%	+12.1%	+31.2%	+18.4%	+12.7%
2022	-11.4%	-0.9%	-11.5%	-18.4%	-11.5%	-13.9%
2023	+19.6%	+8.6%	+4.5%	+7.2%	+2.9%	+18.9%
2024	+12.8%	+2.5%	+2.8%	+3.7%	-4.0%	+2.0%
2025	+3.7%	+28.2%	+16.4%	+23.8%	+31.1%	+23.6%
2026	+10.2%	+4.1%	+7.2%	+10.8%	+13.7%	+8.6%

7. Trading record

The rebalance log per universe is in `results/xs_FS07_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	51 / 50	23%	4%	50%	-19.7%	2020-03
EU	163	24 / 23	6%	6%	44%	-9.0%	2020-07
UK	163	25 / 24	13%	4%	53%	-9.6%	2022-01
DK	163	7 / 6	9%	4%	47%	-14.3%	2020-03
SCANDI	163	13 / 12	4%	6%	53%	-9.1%	2024-10
World	163	98 / 97	16%	5%	50%	-11.8%	2020-03

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: smallest names in the universe	Short: largest names
US	AAL, VIAV, APTV, WYNN, PNR, ZION, AYI, FRT	NVDA, AAPL, GOOGL, MSFT, AMZN, AVGO, META, TSLA
EU	NYR.BR, OBEL.BR, AGFB.BR, GTC.WA, KER.WA, TESB.BR, BPOST.BR, ONTEX.BR	NOVO-B.CO, ASML.AS, VOLV-B.ST, INVE-B.ST, ERIC-B.ST, SAAB-B.ST, AZN.ST, ATCO-A.ST
UK	RTW.L, BPCR.L, PEY.L, MNTN.L, MTLN.L, HANA.L, PRN.L, NAS.L	AZN.L, HSBA.L, RR.L, SHEL.L, GLEN.L, AAL.L, RIO.L, BP.L

Universe	Long: smallest names in the universe	Short: largest names
DK	NDA-DK.CO, BAVA.CO, GN.CO, RBREW.CO, AMBU-B.CO, ISS.CO, MAERSK-A.CO, JYSK.CO	NOVO-B.CO, DSV.CO, VWS.CO, MAERSK-B.CO, DANSKE.CO, NSIS-B.CO, GMAB.CO, PNDORA.CO
SCANDI	KALMAR.HE, QTCOM.HE, KEMIRA.HE, HIAB.HE, TYRES.HE, TIETO.HE, MANTA.HE, HUH1V.HE	NOVO-B.CO, VOLV-B.ST, INVE-B.ST, ERIC-B.ST, SAAB-B.ST, AZN.ST, ATCO-A.ST, DSV.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | |---|---|---|---|---|---| | US | JKP USA 7 themes | -1.6% | -0.78 | 0.59 | mkt +0.25 (+3.6), size +0.43 (+3.2), value +0.44 (+4.3), short_term_reversal +0.44 (+2.2) | | EU | French Europe 5F + WML | -2.0% | -0.64 | 0.21 | SMB +0.49 (+3.7), HML +0.36 (+2.2) | | UK | JKP GBR 7 themes | +6.4% | 2.07 | 0.39 | size +0.94 (+6.9), value -0.53 (-3.1), momentum -0.23 (-2.0), quality +0.41 (+2.1) | | DK | JKP DNK 7 themes | -5.4% | -1.35 | 0.15 | size +0.49 (+3.0), low_risk +0.32 (+2.1) | | SCANDI | French Europe 5F + WML | +1.7% | 0.70 | 0.06 | none |t| ≥ 2 | | World | JKP World 7 themes | -4.1% | -2.06 | 0.44 | mkt +0.15 (+4.2), size +0.52 (+4.3), value +0.27 (+3.0), short_term_reversal +0.26 (+2.2) |

Long-only

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | |---|---|---|---|---|---| | US | JKP USA 7 themes | +3.3% | 2.47 | 0.93 | mkt +1.10 (+23.6), value +0.38 (+4.1), quality +0.24 (+2.1), short_term_reversal +0.47 (+2.8) | | EU | French Europe 5F + WML | +4.4% | 1.73 | 0.64 | Mkt-RF +0.71 (+10.5), HML +0.39 (+2.3) | | UK | JKP GBR 7 themes | +13.1% | 4.34 | 0.69 | mkt +0.51 (+7.6), low_risk -0.49 (-3.0) | | DK | JKP DNK 7 themes | +2.7% | 0.69 | 0.62 | mkt +0.76 (+9.9), value +0.27 (+2.9), low_risk -0.28 (-2.6) | | SCANDI | French Europe 5F + WML | +9.1% | 3.01 | 0.51 | Mkt-RF +0.72 (+8.8) | | World | JKP World 7 themes | +4.7% | 1.74 | 0.83 | mkt +0.80 (+14.4), value +0.33 (+3.1) |

9. Statistical verdict

- **Gate:** |t| > 2.87 (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 4 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -5.5% to +2.9%, |t| past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 1 of the six; alpha -2.6% to +5.8%, passing in 1 of 6.
- **Probabilistic Sharpe** of the L/S, P(true Sharpe > 0): US 0.55, EU 0.26, UK 0.95, DK 0.22, SCANDI 0.82, World 0.53.
- **Publication decay:** gross L/S -3.8% to +18.6% a year in 2013–19 against -7.9% to -1.0% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+4.4%	0.25	+3.5%	+0.9%	-0.7%	+3.6%	+0.8%	30%	2015-07 -5%, 2015-09 -4%, 2014-09 -4%
EU	-1.9%	-0.01	-0.1%	-1.8%	-0.3%	-0.6%	-1.2%	13%	2014-05 -6%, 2019-02 -4%, 2014-10 -4%
UK	+18.6%	-0.10	-1.4%	+19.9%	-0.4%	+11.4%	+7.2%	16%	2016-06 -6%, 2019-03 -4%, 2016-09 -3%
DK	-3.8%	-0.02	-0.3%	-3.5%	-0.3%	-3.2%	-0.6%	14%	2014-02 -9%, 2019-05 -9%, 2015-02 -8%
SCANDI	+7.3%	0.05	+0.8%	+6.5%	-0.3%	+2.9%	+4.3%	11%	2017-02 -4%, 2018-09 -4%, 2019-04 -4%
World	+5.2%	0.17	+2.2%	+3.0%	-0.5%	+2.6%	+2.6%	22%	2016-06 -3%, 2018-09 -2%, 2014-07 -2%

On average across the six universes the gross spread was +5.0% a year, of which the market exposure (beta +0.06) contributed +0.8%; the beta-adjusted spread (CAPM alpha) was +4.2%. Costs took 0.4% a year at 17% monthly turnover across both legs. The long leg beat the universe by +2.8% and the short leg lagged it by +2.2% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages -1.3%, -1.9%, -2.1% a year.

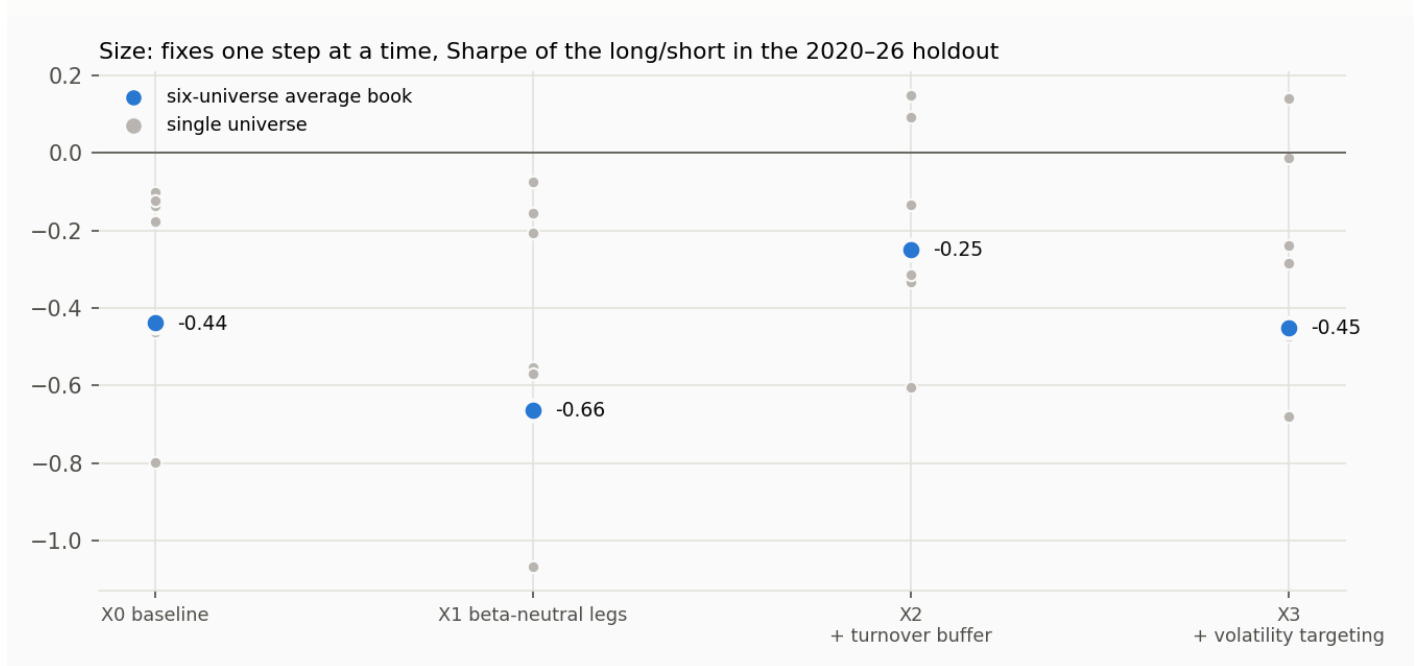
10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.55	–	-0.44	–	–	-3.4%	-28%	-6.2% (-2.2)	0.4%

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X1 beta-neutral legs	0.65	+0.38 (p 0.029)	-0.66	-0.23 (p 0.979)	1 of 6	-5.4%	-35%	-7.6% (-2.4)	0.5%
X2 + turnover buffer	0.69	+0.06 (p 0.379)	-0.25	+0.41 (p 0.061)	5 of 6	-1.4%	-17%	-2.6% (-1.2)	0.2%
X3 + volatility targeting	0.63	-0.22 (p 0.718)	-0.45	-0.20 (p 0.963)	1 of 6	-3.7%	-24%	-5.9% (-1.7)	0.1%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



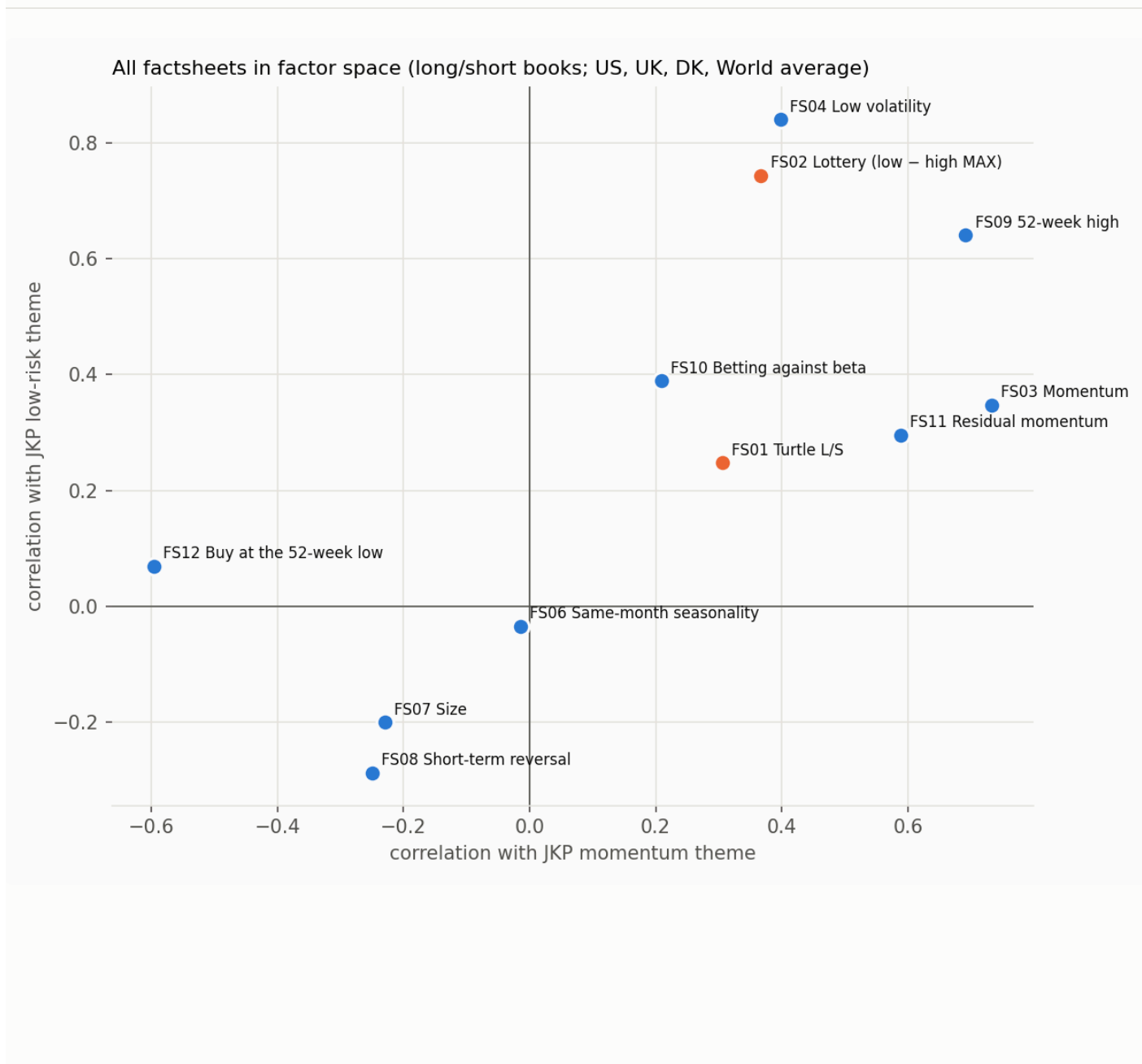
- **X1 beta-neutral legs:** +0.38 design, -0.23 holdout; helps in one window and hurts in the other: not reliable.
- **X2 + turnover buffer:** +0.06 design, +0.41 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.22 design, -0.20 holdout; hurts in both windows.

Where it ends: holdout Sharpe -0.44 → -0.45, return -3.4% → -3.7% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Size
Series family	Size
Academic style	Size premium (Fama-French SMB)
Signal	Price only (daily total returns)

Dimension	Size
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 17% of the two legs replaced monthly
Market exposure	L/S beta 0.08 to 0.41
Payoff shape	Treynor–Mazuy γ -2.04 to 1.84 (t -2.0 to 1.1); worst 10% of market months -2.5% to -0.4% a month, best 10% -0.5% to +2.6%
Nearest factor theme	mkt (2 of 6 universes), HML (1 of 6 universes)
Nearest library signals (returns)	Near 52-week high (3 universes), 55-day breakout (3 universes), Momentum 12-1 (2 universes)
Economic rationale	Risk (distress, illiquidity; Fama & French 1993) or mispricing; much of the historical premium came from the smallest, least liquid stocks and January.
Publication	Banz 1981; Fama & French 1992–93; 'Is size dead?' (van Dijk 2011); 'Size matters, if you control your junk' (Asness et al. 2018)
Where it fits	In the momentum cluster of the library; fully explained by its neighbours.

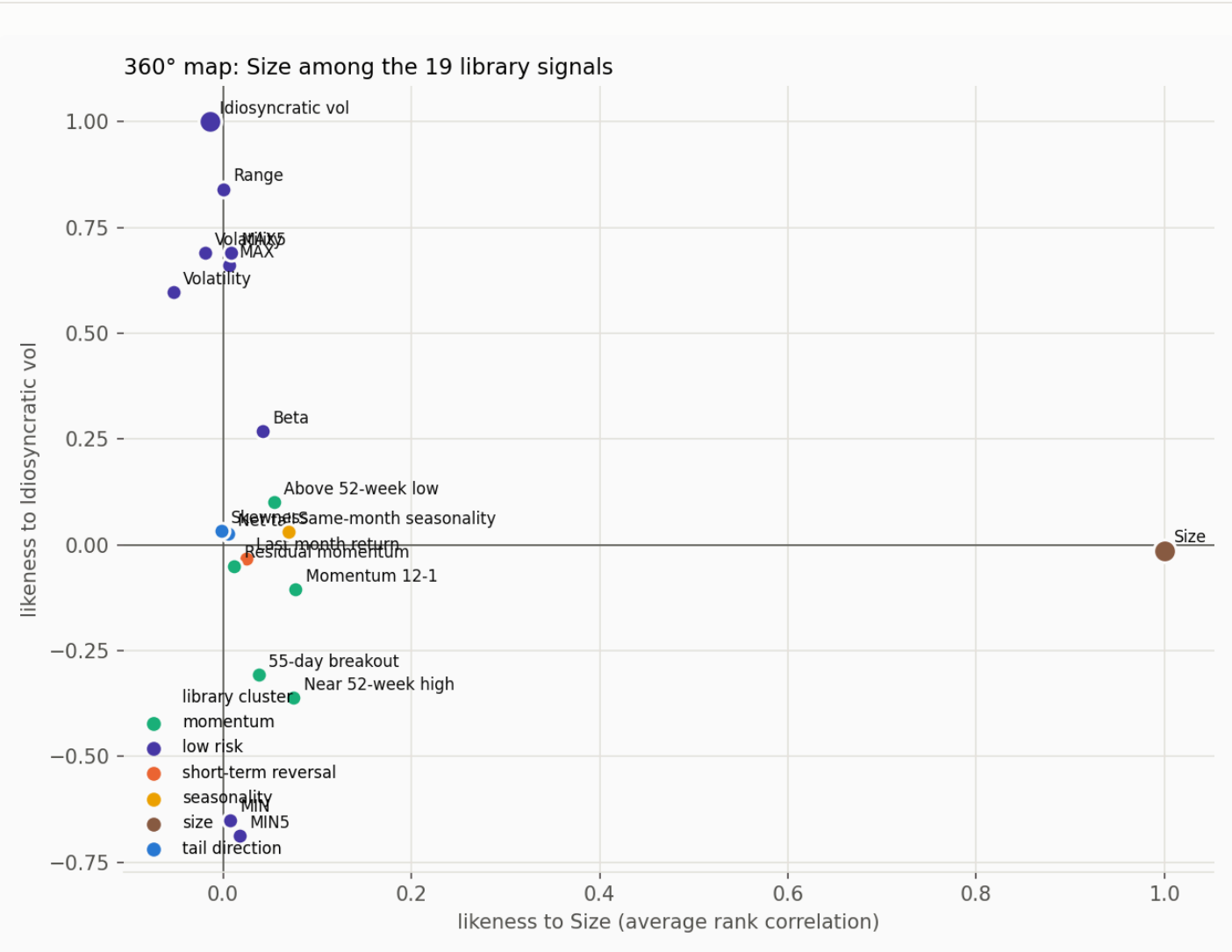


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Idiosyncratic vol.**

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Size (this strategy)	size	+1.00	+1.00	-1.5%	-3.8% to +4.6%	0 / 0
Near 52-week high	momentum	+0.07	+0.31	+2.5%	+8.3% to +18.8%	6 / 0
55-day breakout	momentum	+0.04	+0.27	+0.7%	+5.6% to +12.6%	3 / 0
Momentum 12-1	momentum	+0.08	+0.25	+7.7%	+3.9% to +17.8%	5 / 0
Volatility (252 days)	low risk	-0.05	-0.24	+5.1%	-10.3% to -0.2%	0 / 1
MIN5	low risk	+0.02	+0.23	-2.0%	+2.4% to +13.2%	4 / 0
MIN (worst day)	low risk	+0.01	+0.22	-1.5%	+2.9% to +11.0%	3 / 0
Volatility	low risk	-0.02	-0.22	+3.0%	-10.3% to -3.7%	0 / 5
Last month return	short-term reversal	+0.02	+0.21	+0.9%	-3.2% to +8.4%	0 / 0
Beta	low risk	+0.04	-0.21	+5.5%	-15.7% to -2.6%	0 / 3
Range (MAX–MIN)	low risk	+0.00	-0.20	+1.2%	-11.3% to -2.3%	0 / 4
Idiosyncratic vol (mirror)	low risk	-0.01	-0.19	-1.5%	-10.4% to -0.8%	0 / 3
MAX (best day)	low risk	+0.01	-0.14	+1.9%	-9.3% to +1.0%	0 / 2
MAX5	low risk	+0.01	-0.14	+2.6%	-8.5% to +0.4%	0 / 2
Above 52-week low	momentum	+0.05	+0.14	+8.8%	+4.4% to +13.8%	4 / 0
Net tail (MAX+MIN)	tail direction	+0.01	+0.12	-1.0%	-5.3% to +4.2%	0 / 0
Residual momentum	momentum	+0.01	+0.11	+6.1%	+7.3% to +13.0%	5 / 0
Same-month seasonality	seasonality	+0.07	+0.02	-1.8%	-8.5% to +7.2%	0 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Skewness	tail direction	-0.00	-0.01	-0.2%	-5.4% to +4.5%	0 / 1



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t | R² | Neighbours with |t| ≥ 2 | ---|---|---|---|---| | US | +0.8% | 0.32 | 0.53 |
Momentum 12-1 +0.23 (+3.6) | | EU | -1.5% | -0.47 | 0.15 | | | UK | -2.1% | -0.61 | 0.12 | Same-month seasonality -0.15 (-3.1) |
| DK | +1.6% | 0.45 | 0.15 | Momentum 12-1 +0.29 (+2.1), Residual momentum -0.15 (-2.0) | | SCANDI | +0.9% | 0.32 | 0.12 |
Net tail +0.19 (+2.3), Same-month seasonality -0.20 (-2.4) | | World | +1.1% | 0.58 | 0.43 | |

3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Size	12.8%	13.7%	13.3%
middle	11.5%	13.4%	12.5%
high Size	12.2%	12.1%	10.9%
	low Idiosyncratic vol	middle	high Idiosyncratic vol

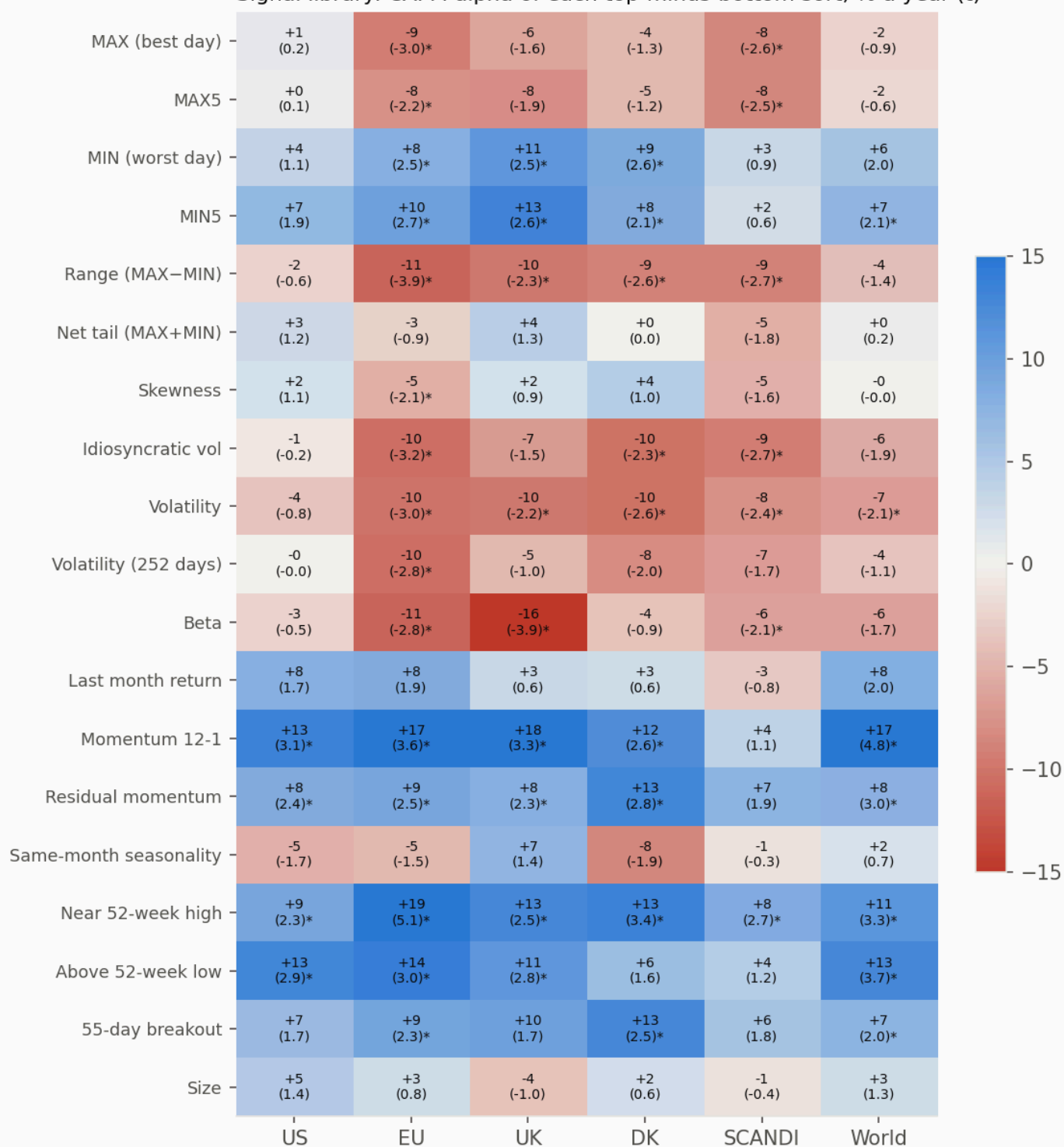
Holding Idiosyncratic vol fixed (down a column), moving from low to high Size changes the return by -0.6, -1.6 and -2.4 points a year; holding Size fixed (along a row), moving from low to high Idiosyncratic vol changes it by +0.5, +1.0 and -1.3.

Market direction. Top-minus-bottom return in rising vs falling market months (six-universe average): -6% vs +7% a year.

Horizon: the gross spread at 1, 2, 3 and 6 months after formation averages -1.5%, -1.3%, -1.9%, -2.1% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Data gap.** No shares outstanding outside the US: size is proxied by traded value, which also measures liquidity and turnover.
5. **Data gap.** The universes are blue chips; the classic size premium lives in micro and small caps, which are excluded by construction.
6. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

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15. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse_xs.py FS07 → build_xs.py
FS07. Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.